

A Packing of 17 Four-Spaces in $\mathbb{R}^9$

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Abstract

We give 17 4-dimensional subspaces of $\mathbb{R}^9$ whose smallest squared chordal distance is at least 2.360730. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.352615, obtained by deleting 1 of the 18 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 17 subspaces of dimension 4 in $\mathbb{R}^9$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.360730 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 9$, $n = 4$. The cell $N = 17$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.352615, obtained by deleting 1 of the 18 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(9 - 4)/9 \cdot 17/16 = 2.361111$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m9n4N17.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 63 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

590182676739332972235062717387661020817365279337451971701746105118875356
262588121379820992391750779792067282540226834238937700736425585792077896
518755133141291260521683853097011924089807422885

denominator:

250000000000220971654833853360793817216595174219089465218338674362144078
738857808380658721298567536672249078545718416180764788235167505189422960
809915798556973451486711990963855376052544991616

B Coordinates

17 blocks of 9 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m9n4N17.txt`.

```
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